Qingbin Li

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EDUCATION

Shanghai Jiao Tong University Sep. 2023 – Jun. 2026

- M.Eng. in Electronic Science and Technology

Xi'an Jiaotong University Sep. 2018 – Jun. 2022

- B.Eng. in Information Engineering

PUBLICATION

- Y. Li, R. Wang, Y. Mei, X. Sui, X. Luo, Z. Wang, H. Shi, **Q. Li**, W. Sun, C. Zhang *et al*, " A 129-Gb/s CMOS W-band Polarization-Modulated Active Relay Transceiver Supporting Concurrent Dual-Polarized Dual-Stream Relay Operation," in 2026 Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits). (accepted)
- W. Sun, Y. Li, Z. Wang, X. Sui, X. Luo, Y. Mei, **Q. Li**, R. Wang, H. Shi and J. Pang, "RF-FilterLLM: A Fine-Tuned LLM Agent for Automated RF Filter Design," in 2026 International Conference on Microwave and Millimeter Wave Technology (ICMMT). (accepted)
- Z. Wang, Y. Li, Y Mei, X. Sui, X. Luo, R. Wang, **Q. Li**, Z. Feng and J. Pang, "A Broadband K/Ka-Band Transformer-Based Three Tline Parallel Doherty Power Amplifier in 65nm CMOS," in 2026 International Conference on Microwave and Millimeter Wave Technology (ICMMT). (accepted)
- **Q. Li**, Y. Li, Z. Wang, X. Luo, R. Wang, Y. Mei, X. Sui, Z. Feng, L. Wu, J. Mao and J. Pang, "A 0.14mm^2 Mixed-Type True-Time Delay Circuit Based on Slow-Wave Transmission Lines Covering 52.8ps for 5G/B5G Carrier Aggregation," in 2026 IEEE MTT-S International Microwave Symposium (IMS). (accepted)
- Z. Wang, Y. Li, Y. Mei, X. Sui, **Q. Li**, X. Luo, R. Wang, D. Ni and J. Pang, "A 24-GHz CMOS Transformer-Based Three-Tline Series Doherty Power Amplifier Achiving 39% PAE," in 2025 IEEE International Conference on Integrated Circuits, Technologies and Applications (ICTA), 2025, pp. 13–14.
- **Q. Li** and J. Pang, "An Area-Efficient 20-100-GHz Phase-Invariant Switch-Type Attenuator Achieving 0.1-dB Tuning Step in 65-nm CMOS," in 2025 18th IEEE United Conference on Millimeter Waves and Terahertz Technologies (UCMMT), vol. Volume1, 2025, pp. 1–3.

RESEARCH EXPERIENCE

RF Circuit Optimization Based on Bayesian Optimization and Gaussian Process (Leader) Aug. 2025 – Dec. 2025

- Introduce a multi-objective Bayesian optimization (MOBO) framework to systematically approximate the performance upper bound under a limited sample budget, enabling comprehensive multi-metric optimization for RF circuit design.

- Utilize Latin Hypercube Sampling (LHS) to generate initial design points across the entire design space, and evaluate their performance using circuit simulators. Then use the collected data to construct an initial Gaussian Process (GP) surrogate model.
- Integrated the Expected Hypervolume Improvement (EHVI) acquisition function into multi-objective Bayesian optimization to efficiently approximate the Pareto front.

Mixed-Type True-Time-Delay (TTD) Circuit for Wideband 5G/B5G Phased-Array Systems (Leader) Oct. 2024 – Jun. 2025

- Applied TTD in wideband systems to mitigate the beam squint issue. Cascaded a reflection-type TTD with two switch-type TTDs in the mixed-type TTD to achieve a wide delay range and fine tuning resolution.
- Implemented the reflection-type TTD using an ultra-compact two-turn 90° coupler and tunable reflective loads. Implemented a switched TTD based on an all-pass network.
- Used slow-wave transmission lines (SW-TLs) to improve delay efficiency per unit area.

A Compact Low-Phase-Error Wideband Attenuator for mm-Wave Phased-Array Systems (Leader) Mar. 2024 – Sep. 2024

- Implemented an attenuator chip based on the T-type structure in a 65-nm CMOS process.
- Added capacitor to the shunt branch of the T-type structure in the large attenuation unit to reduce phase error between the reference and attenuation states.
- Adopted a simplified T-type structure in the small attenuation unit. Implemented resistors in the small attenuation unit using metal lines to reduce parasitic capacitance, thereby minimizing amplitude and phase errors and improving operating bandwidth.

AWARD

- Best Student Paper Award Honorable Mention, 18th IEEE United Conference on Millimeter Waves and Terahertz Technologies(UCMMT), 2025.
- Champion, the "Fellow Townsmen Cup" Basketball Tournament, Xi’an Jiaotong University, 2022.

SKILL

Softwares: Python, PyTorch, ADS, Cadence, HFSS, Origin, Microsoft Office.
Languages: Mandarin (native), English
Hobbies: Basketball, Football, Badminton